

EXPT 3 :PERFORMANCE ANALYSIS OF IIR FILTERS

3.1 DESIGN AND COMPARATIVE ANALYSIS OF BUTTERWORTH AND CHEBYSHEV TYPE-I IIR BAND-PASS FILTERS USING MATLAB

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
% Specifications
```

```
Fs = 2000;
```

```
Rp = 1;
```

```
Rs = 40;
```

```
Wp = [300 600]/(Fs/2);
```

```
Ws = [200 700]/(Fs/2);
```

```
% Butterworth
```

```
[n1,W1] = buttord(Wp,Ws,Rp,Rs);
```

```
[b1,a1] = butter(n1,W1,'bandpass');
```

```
% Chebyshev Type-I
```

```
[n2,W2] = cheb1ord(Wp,Ws,Rp,Rs);
```

```
[b2,a2] = cheby1(n2,Rp,W2,'bandpass');
```

```
%% Magnitude Response
```

```
figure;
```

```
freqz(b1,a1,1024,Fs);
```

```
title('Butterworth Bandpass Filter');
```

```
figure;
```

```
freqz(b2,a2,1024,Fs);  
title('Chebyshev Type-I Bandpass Filter');
```

```
%% Group Delay  
figure;  
grpdelay(b1,a1,1024,Fs);  
hold on;  
grpdelay(b2,a2,1024,Fs);  
legend('Butterworth','Chebyshev-I');  
title('Group Delay Comparison');
```

```
%% Pole-Zero Plot  
figure;  
subplot(1,2,1);  
zplane(b1,a1);  
title('Butterworth Pole-Zero Plot');
```

```
subplot(1,2,2);  
zplane(b2,a2);  
title('Chebyshev-I Pole-Zero Plot');
```

```
%% Impulse Response  
figure;  
subplot(2,1,1);  
impz(b1,a1);  
title('Butterworth Impulse Response');
```

```
subplot(2,1,2);  
impz(b2,a2);  
title('Chebyshev-I Impulse Response');
```

```
%% Pole Magnitudes

figure;
subplot(1,2,1);
stem(abs(roots(a1)), 'filled');
grid on;
title('Butterworth Pole Magnitudes');
xlabel('Pole Number');
ylabel('|Pole|');

subplot(1,2,2);
stem(abs(roots(a2)), 'filled');
grid on;
title('Chebyshev-I Pole Magnitudes');
xlabel('Pole Number');
ylabel('|Pole|');

%% Display Filter Orders
fprintf('Butterworth Filter Order = %d\n', n1);
fprintf('Chebyshev Type-I Filter Order = %d\n', n2);
```







